Table 1: Classical computing vs quantum computing

Dimension	Classical computing	Quantum computing
Governing physical law	Electromagnetism governing electric currents; Classical electrodynamics	Quantum mechanics governs sub-atomic particles; Quantum electrodynamics
Primary scaling principle	Moore's Law	Harnesses the power of sub-atoms and molecules to perform memory and processing tasks
Fundamental information unit	Digital bits 0,1	Quantum bits (qubits), superposition of 0 and 1
Operational state model	Deterministic	Stochastic
State capacity (units)	n-bits: 2n sequences available, but only one used at a time	n-qbits: 2n sequences available in parallel
Processing paradigm	Sequential computing	Parallel computing
Computational speed metrics	One operation at a time	Millions of operations at a time
Relative throughput (example)	64-bit classical computer operates at speeds measured in gigaflops (billions of floating-point operations per second)	30-qubit quantum computer equals the processing power of conventional computers that run at 10 teraflops (trillions of floating-point operations per second)

remotely on the cloud. Only a limited portion of the programming tool stack is deployed on local machines. Programmers access production-ready quantum software remotely for development and testing.

Quantum algorithms: Programmers leverage quantum algorithms to define the computing task. Algorithms help in gaining speed and communicating with other computing tasks that are running on QCPs. Programmers must also identify or design suitable quantum algorithms that can solve the problem on hand.

Portability of software: Quantum computing platforms are currently evolving and the software developed by the platform owners are native in nature. This software follows their own standards, with proprietary programming APIs and tools.

QCP architecture

QCP architecture consists of an end-to-end quantum computing system with seven layers, starting with the physical qubit at the bottom and progressing upwards to the application/business layer (Figure 1). Quantum computation is a tightly coupled system of quantum hardware, cryogenic infrastructure, control electronics, real-time software, compilers, orchestration, and classical computing.

QPU and physical qubits: The quantum execution layer consists of quantum information physically existing in quantum gates, interactions and measurements.

Cryogenic environment and interconnect: This layer provides the physical environment and signal path between room-temperature electronics and the QPU. It is essentially the infrastructure that allows the QPU to live at millikelvin temperatures while still being controlled from the classical environment.

Control and readout electronics: This layer converts digital instructions into physical electrical/microwave signals that manipulate the qubits. It also performs the reverse operation: converting very weak signals coming from the QPU into usable digital data.

Quantum classical interface/ control highway: This acts as a quantum computer's high-speed, thermally managed nervous system. It is the boundary and transport mechanism between classical electronics and the quantum device.

Classical control and quantum orchestration: This layer is the real-time brain coordinating the quantum machine. It sits between the quantum software stack and the physical control system.

Quantum software, compiler and runtime: This layer translates human/ programmer intent into executable quantum operations.

Application and business layer: This is the layer where business value is created. It leverages quantum software applications that help in meeting the business requirements of an enterprise. Main application areas are:

- *Optimisation:* Supply-chain optimisation, scheduling, portfolio optimisation, route optimisation
- *Chemistry:* Molecular simulation, drug discovery, catalyst design
- *Materials:* Materials simulation, battery materials, superconducting materials
- *Finance:* Portfolio optimisation, risk analysis, pricing
- *Cybersecurity:* Cryptographic research, security analysis